BUSINESS DATA MANAGEMENT

Lecture06

Demand Elasticity

Hello friends, in this particular video we will try to understand in depth what we mean when we say elasticity of demand or demand elasticity. So let's get started quickly with the concept. So when somebody says hey, the demand of this particular product is elastic or it is inelastic, what does that really mean? It means that what is the degree of the responsiveness of demand with respect to any change in price.

So very fundamentally speaking, when we say anything to do with elasticity, it is nothing but trying to understand the degree of responsiveness of change in one variable with respect to the other variable. In the context of demand elasticity, we are trying to understand what is the change in demand when there is a change in price? How does it impact? To what extent does it impact? So when we say a demand is considered as elastic or highly elastic, it means that the quantity demanded responds substantially.

There is a big change in the demand when there is a change in price. On the other hand, when we say the demand is inelastic or less elastic, it means that this particular change is far less to a far lesser degree as compared to the change in price. Let's take an example of Netflix. Let's say tomorrow Netflix raises prices and it becomes unaffordable for many consumers, customers, and they cancel the subscription.

What would you call it? Would you call this demand elastic or inelastic? It means that the demand for Netflix is elastic. Think about things which are very critical. Let's say again, given the geopolitical situation, there has been an increase in petrol prices.

Do you think the demand will substantially come down because there is an increase in prices? The answer is not really. And therefore, for this particular

product or good which is petrol, the demand is regarded as inelastic. So what determines elasticity? What are the determinants of elasticity of demand?

While there are no universal rules, there are some basic thumb rule principles that you should be looking at when you are trying to analyze the elasticity of the good or the services that you are analyzing as a data analyst. For example, you should be looking at what's the competition like? What is available in the market? If for your good or your product or service that you are analyzing, if you see that there are very close substitutes, that means the elasticity here will be higher. Why?

Because the consumers would easily be able to switch from one product, which is yours, to a competing product. But let's say your product is unique. Or it's highly differentiated. In that case, this whole elasticity will not be that high. Taking a very simple example, if you look at, let's say, something like butter, there are substitutes to butter.

And if, let's say, the price of butter goes up, you could use other substitutes to butter. But let's say there is a good, like, egg. Eggs, there is an increase in price of eggs. Are there really a lot of substitutes for eggs? Not really.

So in this case, the demand elasticity for eggs would be regarded as less elastic. The second thing, and you would have probably got it from the example that we discussed in the previous slide on Netflix. You should look at what is the type of good that you're talking about. Is it a necessity or is it more of a luxury? Because if the product or the services are critical or is a necessity, in that case, the demand would typically be inelastic.

Whereas, for products, goods or services that are good to have, nice to have, luxurious products, the demand would be highly elastic. Okay, take an example of a service of a doctor. Do you think the doctor's visit is a necessity or a luxury? It's a necessity and therefore they will tend to have an inelastic demand versus let's say the demand for luxurious products, luxurious goods, luxurious services. In this case, the demand would be highly elastic.

So the type of good that we're talking about also has a big bearing on the elasticity of that particular good with respect to its demand if there is a change in price points, for instance. The second is to look at which market is my product or service being catered to. If you have a very narrowly defined market, you have a more elastic demand than a more broadly defined market. So if you're looking at a specific good here, which is let's say a specific ice cream here, a vanilla ice cream here, it's a very narrowly defined product that you're talking about, your demand would be more elastic as compared to if let's say you're in the food business and you are having a variety of goods here to be catered to. Now, because of that, the definition of market that you are looking at over here, which is far more broadly defined,

this will become more inelastic. So a broadly defined market, because it is easier to find close substitutes in the case of narrowly defined goods, for a broadly defined market it is not there and therefore it will be more inelastic. Now it's also important to understand that there is a whole time horizon concept here. Goods will tend to have more elastic demand over a longer period of time horizons. For example, if you're looking at change in prices of petrol because there's lack of substitutes, the demand will immediately not

Fall down, but over a period of time, the demand will move towards products that may be more substitutable to a petrol. For example, there could be more demand for electric goods. Electric vehicles, other kinds. So the shift will be for a longer time horizon compared to more elastic oriented goods where the shift can be far more quickly done. So, for example, in case of normal goods where elasticity is high, the shift will be in a shorter period of time, whereas in case of relatively less elastic goods, the shift will happen, but over a longer period of time.

It's important to really understand what is the time horizon of my good, my service that I'm catering to where there will be a shift in demand with a shift in preferences, a shift in pricing. Now, how do you go about computing elasticity of demand? We know that it's the change in demand with respect to the change in price. So if you look at it from a quantitative mathematical perspective, you will

see the change in demand is in terms of what's the percentage change in demand versus the percentage change in price. So when you look at these two as a ratio, which is percentage change in quantity demanded versus percentage change in price, what you will get is an indication of the level of elasticity.

For example, if suppose there is a 10% increase in the price of an ice cream cone and the amount of ice cream you buy falls by 20%, so the quantity demanded falls by 20%, Whereas the price only increased by 10%. So now if you look at it from a ratio perspective, you will see 20% was the change in quantity demanded versus 10% was the fall in pricing. So the ratio is two, which means this particular good is highly elastic. Okay.

So. Simply put, you know, before we get on to the classification table, if the ratio is higher than one, that means the good is elastic; if the ratio is less than one, that means this particular good is inelastic; the demand is inelastic for that particular good because the shift in quantity is not as high as as the shift in price point because it's a necessity, because it is critical. So again, as a data analyst, it's important for you to again keep a reference check to what is the kind of good or service you're catering to, what is the kind of differentiation that you have, what is the kind of market you are catering to, at what price point. Just to summarize here, we discussed about elasticity, so what are the different types of elasticities of demand. So as we discussed, you have an elastic demand, which means an elastic demand means a small change in price leads to a relatively larger change in the quantity demanded for that particular good.

Example could be restaurants, clothes, branded clothes, specifically branded clothes. Why? Because they're good to have. So here, if there's a change in price, there's a significant shift in the quantity demanded. Whereas in case of inelastic demand, the change in price leads to a relatively smaller change in quantity demanded, sometimes even none.

We'll talk about that. So those kinds of goods where there's a change in price does not really impact the quantity demanded, that's called as inelastic demand

goods. For example, medicines. For example, petrol. Unit elastic demand is, I would say, a theoretical concept here.

The change is exactly the same. A 10% fall in price leads to a 10% rise in demand. If that happens, it's called unit elastic. I would still say it's elastic, but it is unit elastic demand. The proportionate change remains largely the same.

A perfectly elastic demand means that even a tiny increase in price causes the demand to fall significantly. This is for commoditized markets. A high number of substitutes are there. It's a perfectly competitive market.

A number of players are there. So if you change the price, you move the price from 10 to 12, there is a huge downward spiral in the demand to zero because the consumers can move, they can move from your good to another substitute good. And in the same way, a perfectly inelastic demand is where the demand remains largely constant regardless of the change in price.

So these are very critical products we're talking about. We're talking about significantly important, necessary products, goods or services without which it would be difficult to survive. For example, life-saving insulin for patients who are diabetic. To be honest, the demand really does not change at all irrespective of the change in price points.

So, again there are five types of demand elasticity. You have elastic demand, inelastic demand, unit elastic demand, perfectly elastic demand and perfectly inelastic demand. Let's look at this in a graphical representation, what we just discussed. If you look at the figures on the right hand side, let's go with unit elastic demand. Let's look at the y-axis and all of these graphs is the price point and the x-axis is the quantity demanded.

So what happens when the price point moves from four to five point something? There's a 20 plus percent, a 22% change in price point here. What happens to the quantity? The quantity also changes exactly by the same percentage point

from 100 to something close to 80, which makes it 22%. So both of them change by the equal percentages, and that's why it's called unit elastic.

Demand. When you look at an elastic demand, which is, if you go down to the left hand side, which is graph number d, you will see that there is a shift in pricing, but the change in the quantity demanded is quite significant. I think it's an elastic product. There are substitutes available. So that is called elastic demand. And then we have the inelastic demand, which, if you look at graph number B, where there is a shift in pricing, does not really change the demand to a very large degree.

The shift in pricing happened to the extent of, let's say, 20, 25%. In pricing, there is a demand move by 5, 10%. That is called an inelastic demand. Okay, and in this case again, if you look at the ratios for elastic demand, it will be more than one; for an elastic demand, the ratio will be less than one; for a unit of perfect or unit elastic demand, it will be exactly equal to one. And then you have the two extremes: you have a perfectly elastic demand on graph number e, which means that in this particular case, any shift in pricing brings down the demand to zero to nil.

And if you look at the perfectly inelastic demand, any shift in pricing does not change the quantity that is demanded. This is for critical goods. So this is like a graphical representation of the same constructs that we discussed in the previous slides. Now let us look at, finally, as a data analyst, what are you trying to gauge? You're trying to gauge how does this elasticity concept impact revenues.

How does this whole price and demand elasticity impact my sales or revenues? Now what is total revenues? Total revenues is the amount of goods sold multiplied by the price of the good. So quantity sold which is Q multiplied by the market price which is P into Q. That is your total revenues. If the demand is elastic, any change, any increase in the price will tend to decrease the total revenue.

Why? Because the price and the total sales move in opposite direction. Your quantity demanded will be significantly lower. compared to the change in price and therefore when you multiply the new price by the new quantity sold P into Q, your total revenues will tend to fall in a rising price situation. Whereas if the demand for your good is differentiated in elastic, any increase in price will tend to increase the revenues.

Why? Because you increase the price, but the shift in demand was not significantly downwards. It was probably, but little less or maybe the same. Therefore, your total revenues, P multiplied by Q, increases. and in a state when there is unit elastic situation, it will remain the same as we discussed, right?

So it's important again as a data analyst to figure out where does your product lie compared to competitors, compared to the need of the market? How are you differentiating your products? and services, and therefore, how will the impact of any change in price point, whether driven by increase in input prices or whether driven by competitive scenarios, how will it change the demand and therefore, the revenues of your firm? So I hope this was useful and informative. and an interesting lecture video on elasticity and the concept of elasticity.

I hope to see all of you in the next video where we will discuss elasticity in the context of supply. So we discussed the demand elasticity. We will in the next video discuss supply elasticity. So thank you very much and bye for now.